

$$1. G_p(s) = \frac{K}{s+6}$$

$$a) K=1, G(s)=?$$

$$G_p(s) = \frac{1}{s+6} = \frac{Y(s)}{U(s)} \Rightarrow sY(s) + 6Y(s) = U(s) \cdot 1 \Rightarrow \frac{dy(t)}{dt} + 6y(t) = u(t)$$

$$u(t) \approx u(kT)$$

$$y(t) \approx y(kT)$$

$$\frac{dy(t)}{dt} \approx \frac{y(kT) - y(k-1)T}{T}$$

$$\left. \begin{array}{l} u(kT) \approx u(kT) \\ y(kT) \approx y(kT) \\ \frac{dy(t)}{dt} \approx \frac{y(kT) - y(k-1)T}{T} \end{array} \right\} \Rightarrow \frac{y(kT) - y(k-1)T}{T} + 6y(kT) = u(kT) \Rightarrow$$

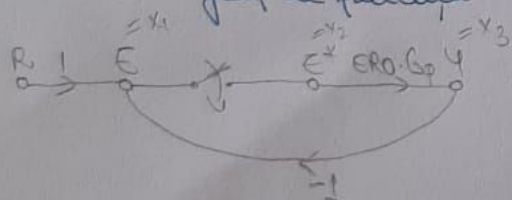
$$\Rightarrow y(kT) - y(k-1)T + 6Te y(kT) = Te u(kT) \Rightarrow$$

$$\Rightarrow y(kT)(1 + 6Te) - y(k-1)T = Te u(kT) \Rightarrow y(2)(1 + 6Te) - 2^{-1} y(2) = Te \cdot u(2) \Rightarrow$$

$$\Rightarrow y(2)(1 + 6Te - 2^{-1}) = Te \cdot u(2) \Rightarrow G_p(z) = \frac{Y(z)}{U(z)} = \frac{Te}{1 + 6Te - 2^{-1}} = \frac{Te z}{z + 6Te z - 1}$$

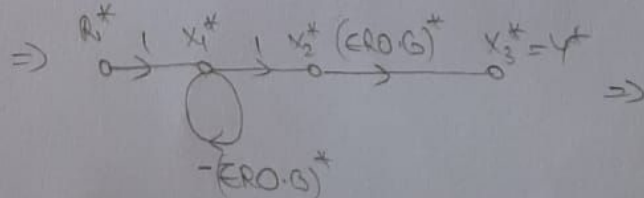
$$G_o(z) = \frac{G_p(z)}{1 + G_p(z)} = \frac{\frac{Te z}{z + 6Te z - 1}}{1 + \frac{Te z}{z + 6Te z - 1}} = \frac{Te z}{z + 6Te z - 1 + Te z} = \frac{Te z}{z(1 + 6Te) - 1}$$

b) $G_o(s)=?$ graf de fluxuri



$$\Rightarrow \begin{cases} x_1 = E = R - Y = R - x_3 \\ x_2 = E^* = x_1^* \\ x_3 = Y = ERO \cdot G \cdot x_2 = ERO \cdot G \cdot x_1^* \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} x_1 = R - ERO \cdot G \cdot x_1^* \\ x_2 = x_1^* \\ x_3 = ERO \cdot G \cdot x_1^* \end{cases} \Rightarrow \begin{cases} x_1^* = R^* - (ERO \cdot G)^* x_1^* \\ x_2^* = x_1^* \\ x_3^* = (ERO \cdot G)^* x_1^* \end{cases}$$



$$\Rightarrow G_o(z) = \frac{(ERO \cdot G(s))^*}{1 - [(ERO \cdot G(s))^*]} \Big|_{s=\frac{1}{T} \ln z} = \frac{z \{ ERO \cdot G(s) \}}{1 + z \{ ERO \cdot G(s) \}}$$

$$z \{ ERO \cdot G(s) \} = z \left\{ \frac{1 - e^{-sT}}{s} \cdot \frac{1}{s+6} \right\} = (1 - z^{-1}) z \left\{ \frac{1}{s(s+6)} \right\} = (1 - z^{-1}) \cdot z \left\{ \frac{c_1}{s} + \frac{c_2}{s+6} \right\} \Rightarrow$$

$$c_1 = \frac{1}{s+6} \Big|_{s=0} \Rightarrow c_1 = \frac{1}{6} \quad ; \quad c_2 = \frac{1}{s} \Big|_{s=-6} \Rightarrow c_2 = -\frac{1}{6}$$

$$G_p(s) = \frac{s-1}{s} \cdot Z \left\{ \frac{1}{s} \cdot \frac{1}{s} - \frac{1}{s} \cdot \frac{1}{s+6} \right\} = \frac{s-1}{s} \cdot \frac{1}{6} \left(\frac{s}{s-1} - \frac{s}{s+6} \right) = \frac{s-1}{6} \cdot \frac{s e^{-6Te} - s + 1}{(s-1)(s+6)} =$$

$$= \frac{1}{6} \frac{1-e^{-6Te}}{s+6}$$

$$\Rightarrow G_0(s) = \frac{1}{6} \frac{\frac{1-e^{-6Te}}{s+6}}{1 + \frac{1-e^{-6Te}}{s+6}} = \frac{1}{6} \frac{1-e^{-6Te}}{s+6+1-e^{-6Te}} = \frac{1}{6} \frac{1-e^{-6Te}}{s-2e^{-6Te}+1}$$

$$= \frac{1}{6} \frac{1-e^{-6Te}}{s-2e^{-6Te}+1} = \frac{1}{6} \frac{1-e^{-6Te}}{s-2e^{-6Te}+1}$$

c) $R(z) = ?$ ist ähnlich.

$$G_0(z) = \frac{1 - e^{-6T_e}}{z - 2 \cdot e^{-6T_e} + 1} \quad \text{mit } T_e = 0,1 \Rightarrow G_0(z) = \frac{0,16}{z - 0,1}$$

$$e_A^* = \lim_{z \rightarrow 1} (1 - z^{-1}) \cdot E(z) = \lim_{z \rightarrow 1} \frac{z-1}{z} \cdot \frac{R(z)}{H(z)}$$

$$R(z) = \frac{z}{z-1} \Rightarrow e_A^* = \lim_{z \rightarrow 1} \frac{z-1}{z} \cdot \frac{z}{z-1} \cdot \frac{1}{1 + 0,16 \frac{1}{z-0,1}} = \frac{1}{1+0,17} = 0,85$$

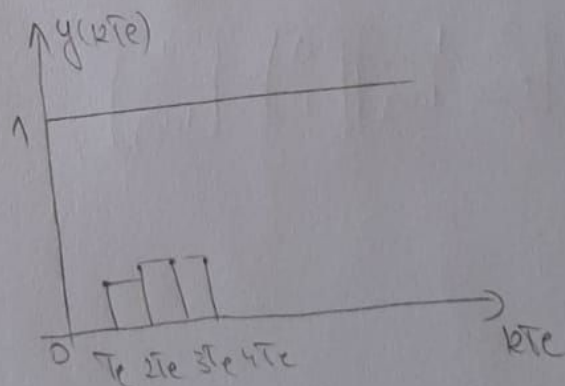
d) $y(kT_e) = ?$

$$y(kT_e) = Z^{-1} \{ Y(z) \} = Z^{-1} \{ R(z) \cdot G_0(z) \}$$

$$Y(z) = \frac{z}{z-1} \cdot \frac{0,16}{z-0,1} = \frac{0,16z}{z^2 - 1,1z + 0,1} \quad | : z^{-2} \Rightarrow Y(z^{-1}) = \frac{0,16z^{-1}}{1 - 1,1z^{-1} + 0,1z^{-2}}$$

$$\begin{array}{r} 0,16z^{-1} \\ - 0,16z^{-1} + 0,17z^{-2} - 0,01z^{-3} \\ \hline 0,17z^{-2} - 0,01z^{-3} \\ - 0,17z^{-2} + 0,18z^{-3} - 0,01z^{-4} \\ \hline 0,17z^{-3} - 0,01z^{-4} \\ - 0,17z^{-3} + 0,18z^{-4} \\ \hline 0,17z^{-4} \end{array}$$

$$y(kT_e) = \{ 0,16; 0,17; 0,17; \dots \}$$



$$2. G_d(z) = \frac{(0.1132 + 0.02z)}{z^2 - 0.39z + 0.007} \quad T_e = 1s.$$

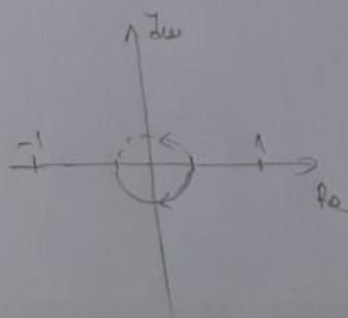
$$\begin{aligned} a) z &= \frac{1 + \frac{T_e}{2}s}{1 - \frac{T_e}{2}s} \Rightarrow z = \frac{1 + 0.5s}{1 - 0.5s} \Rightarrow G_d(s) = \frac{0.113 \cdot \frac{1 + 0.5s}{1 - 0.5s} + 0.02}{\frac{(1 + 0.5s)^2}{(1 - 0.5s)^2} - 0.39 \frac{1 + 0.5s}{1 - 0.5s} + 0.007} = \\ &= \frac{0.113 + 0.07s + 0.02 - 0.02s}{(1 - 0.5s)} \cdot \frac{(1 - 0.5s)^2}{1 + s + 0.25s^2 - 0.39 + 0.07s^2 + 0.07 - 0.09s + 0.01s^2} = \\ &= \frac{(0.15 + 0.05s)(1 - 0.5s)}{0.68 + 0.93s + 0.33s^2} \Rightarrow G_d(j\omega) = G_c(s) \Big|_{s=j\omega} = \frac{(0.15 + 0.05j\omega)(1 - 0.5j\omega)}{0.68 + 0.93j\omega - 0.33\omega^2} = \\ &= \frac{0.15 - 0.07j\omega + 0.05j\omega + 0.02\omega^2}{(0.68 - 0.33\omega^2) + 0.93j\omega} = \frac{0.15 - 0.02j\omega + 0.02\omega^2}{(0.68 - 0.33\omega^2) + 0.93j\omega} = \\ &= \frac{(0.15 - 0.02j\omega + 0.02\omega^2)(0.68 - 0.33\omega^2 - 0.93j\omega)}{0.68 - 0.14\omega^2 + 0.09\omega^4 + 0.86\omega^2} = \\ &= \frac{0.1 - 0.04\omega^2 - 0.13j\omega - 0.01j\omega + 0.009\omega^3 - 0.01\omega^2 + 0.01\omega^2 - 0.006\omega^4 - 0.009j\omega^3}{0.68 + 0.46\omega^2 + 0.09\omega^4} = \\ &= \frac{0.1 - 0.04\omega^2 - 0.06\omega^4 + j(-0.14\omega)}{0.68 + 0.46\omega^2 + 0.09\omega^4} \Rightarrow \operatorname{Re}\{G_d(j\omega)\} = \frac{0.1 - 0.04\omega^2 - 0.06\omega^4}{0.68 + 0.46\omega^2 + 0.09\omega^4} \\ &\quad \operatorname{Im}\{G_d(j\omega)\} = \frac{-0.14\omega}{0.68 + 0.46\omega^2 + 0.09\omega^4} \end{aligned}$$

$$b) n - m = 0.$$

$\omega = 0 \Rightarrow$ p-leacă de pe axa reală din part 0.147 și se termină în -0.067.

$$\operatorname{Im}\{G_d(j\omega)\} = 0 \Rightarrow \omega = 0 \Rightarrow \operatorname{Re}\{G_d(j\omega)\} = \frac{0.1}{0.68} = 0.147$$

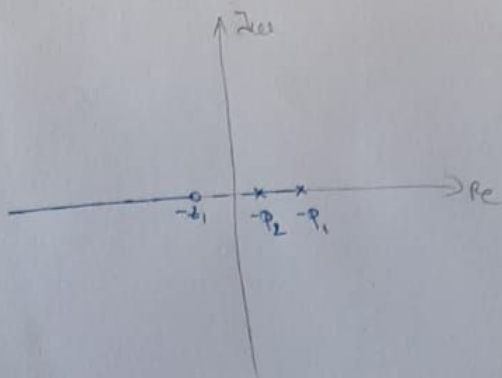
$$\text{dintru } \operatorname{Im}\{G_d(j\omega)\} = 0 \text{ și } \text{dintru } \operatorname{Re}\{G_d(j\omega)\} = \frac{-0.06}{0.9} = -0.067$$



c) nu este în vîră dezechilibru stabil - axa reală nu intersectează -1 j.0.

d) este în vîră dezechilibru stabil

$$3. G(z) = \frac{K(0.16z + 0.04)}{z^2 - 0.4z + 0.02}, T_e = 1s.$$



a). $m = 2$ racines ale doublet reelles.

$m = 1 \Rightarrow u - w = 1$ racine $\rightarrow \infty$.

b) $-p_1 = 0.131 \quad -p_2 = 0.06 \quad j - z_1 = -0.125$

$(-\infty, -0.125] \cup [0.06, 0.131]$.

c) $u - w = 1 \Rightarrow 0$ asymptote

$\sigma_0 = \frac{\pm 180^\circ(2k+1)}{u-w}; k = 0, 1, \dots$

$\sigma_0 = \pm 180^\circ$: une arce pot de instabilitate

d) $1 + \frac{K(0.16z + 0.04)}{z^2 - 0.4z + 0.02} = 0 \Rightarrow K_1 = \frac{-(z^2 - 0.4z + 0.02)}{0.16z + 0.04} \Rightarrow$

$\frac{dK}{dz} = \frac{-(z^2 - 0.4z + 0.02)(0.16z + 0.04) + 0.16(z^2 - 0.4z + 0.02)^2}{(0.16z + 0.04)^2} = 0 \Rightarrow$

$\Rightarrow -0.132z^2 - 0.108z + 0.064z + 0.016 + 0.16z^2 - 0.1084z + 0.0032 = 0 \Rightarrow$

$\Rightarrow -0.16z^2 - 0.08z + 0.048 = 0 \Rightarrow 0.16z^2 + 0.08z - 0.048 = 0 \Rightarrow z_1 = -0.185$
 $z_2 = 0.135$

e) $G(x+yj) = \frac{0.16(x+yj) + 0.04}{(x+yj)^2 - 0.4(x+yj) + 0.02} = \frac{0.16x + 0.16yj + 0.04}{x^2 + 2xyj - y^2 - 0.4x - 0.4yj + 0.02}$
 $= \frac{0.16x + 0.16yj + 0.04}{x^2 - y^2 - 0.4x + 0.02 + 2xyj - 0.4yj}$; $\angle G(x+yj) = 180^\circ(2k+1) \Rightarrow$

$\Rightarrow \arctan \frac{0.16y}{0.16x + 0.04} - \arctan \frac{2xy - 0.4y}{x^2 - y^2 - 0.4x + 0.02} = 180^\circ(2k+1) + \pi \Rightarrow$

$\frac{a-b}{1+ab} = 0 \Rightarrow a = b \Rightarrow \frac{0.16y}{0.16x + 0.04} = \frac{y(2x - 0.4)}{x^2 - y^2 - 0.4x + 0.02} \Rightarrow 0.16x^2 - 0.16y^2 - 0.064x + 0.032 =$
 $= 0.32x - 0.064x + 0.032x - 0.016$

$$0.16x^2 - 0.116y^2 - 0.14x = -0.048$$

$$0.16x^2 - 0.4x + 1.56 - 0.116y^2 = -0.048 + 1.56$$

$$(x + 1.25)^2 - 0.116y^2 = 1.51 \Rightarrow C(-1.25, 0) \quad R = 1.22$$

$$4. G_0(z) = \frac{0,02z + 0,02}{z^2 - 1,49z + 0,54}$$

$$a) G_0(z) = \frac{0,02z + 0,02}{z^2 - 1,49z + 0,54} = \frac{Y(z)}{R(z)} = \frac{Y(z)}{X(z)} \cdot \frac{X(z)}{R(z)} \Rightarrow \frac{Y(z)}{X(z)} = 0,02z + 0,02 \text{ m}$$

$$\frac{Y(z)}{R(z)} = \frac{1}{z^2 - 1,49z + 0,54} \Rightarrow z^2 X(z) - 1,49z X(z) + 0,54 X(z) = R(z) \quad | z^{-1} \Rightarrow$$

$$\Rightarrow X[k+2] - 1,49 X[k+1] + 0,54 X[k] = R[k]$$

$$X[k+1] = X_1[k+1]$$

$$X[k+1] = X_2[k], \quad X[k+1] = X_1[k+1]$$

$$X[k+2] = X_2[k+1] = -0,54 X_1[k] + 1,49 X_2[k] + R[k]$$

$$\Rightarrow \Phi = \begin{bmatrix} 0 & 1 \\ -0,54 & 1,49 \end{bmatrix}; \Gamma = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\frac{Y(z)}{X(z)} = 0,02z + 0,02 \Rightarrow 0,02z X(z) + 0,02 X(z) = Y(z) \quad | z^{-1} \Rightarrow$$

$$\Rightarrow y[k] = 0,02 X[k+1] + 0,02 X[k] \Rightarrow y[k] = 0,02 X_2[k] + 0,02 X_1[k]$$

$$\Rightarrow C = [0,02 \quad 0,02] \text{ m } D = 0.$$

$$b) \Phi = [P \quad \Phi P] = \begin{bmatrix} 0 & 1 \\ 1 & 1,49 \end{bmatrix} \Rightarrow \det P = -1 \neq 0 \Rightarrow \text{nicht kontrollierbar}$$

$$\Phi P = \begin{bmatrix} 0 & 1 \\ -0,54 & 1,49 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1,49 \end{bmatrix}$$